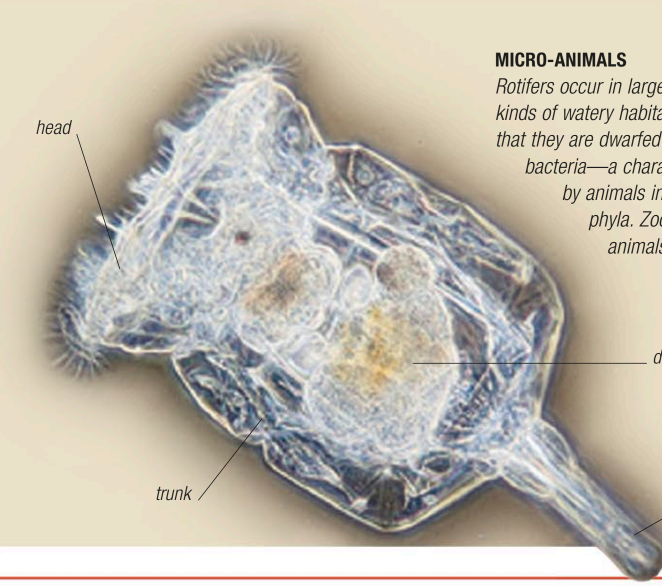
Minor phyla

The invertebrate world is classified into about 30 major groups, or phyla. Some contain common and familiar animals but well over half contain organisms that go largely unnoticed—because they are rare, because they are microscopic, or because they live in inaccessible habitats, such as seabed mud. These animals differ from each other as much as invertebrates differ from vertebrates, and they have widely divergent ways of life. Some minor phyla, such as arrow worms, comprise just a handful of species, while a few—such as bryozoans—contain thousands of species, yet both groups are ecologically important. A selection of minor phyla is shown on these 2 pages.



MICRO-ANIMALS
Rotifers occur in large numbers in all kinds of watery habitats. They are so small that they are dwarfed by some forms of bacteria—a characteristic that is shared by animals in several other minor phyla. Zoologists call these tiny animals micrometazoans.

Phylum Ctenophora

Comb jellies

Occurrence c.200 spp. worldwide; planktonic, a few on the seabed

With their tentacles extended, comb jellies, or sea gooseberries, are some of the most attractive and graceful animals in the sea. Although they can swim, by beating their comblike plates of fine cilia (structures similar to hairs), they make little headway against the current, and are often carried along in large swarms by the tide or the wind. As the cilia rows beat, they produce beautiful, iridescent colors. Most comb jellies are small in size, but some can grow to 6½ ft (2 m) long.

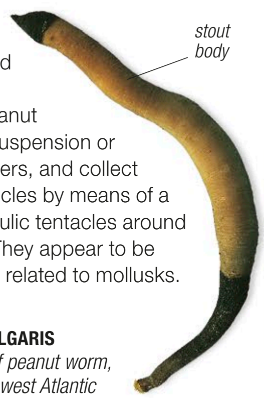
SEA GOOSEBERRY
Comb jellies come in a variety of shapes, from spheres to ribbons and disks. The Atlantic-dwelling Pleurobrachia pileus *(right) is ¾ in (2 cm) long and has a pair of tentacles 6 in (15 cm) long.*

Phylum Sipuncula

Peanut worms

Occurrence c.150 spp. worldwide; in soft sediments, mollusk shells, rock crevices, holes in coral or wood

These plump, unsegmented worms, 1/16–28 in (0.2–70 cm) long, have a peanut- or sausage-shaped body, with walls formed of layers of circular and longitudinal muscles. Their most notable feature is their retractable trunk, known as an introvert, which is extended with the help of hydraulic pressure, and retracted by muscles. Peanut worms are suspension or deposit feeders, and collect organic particles by means of a ring of hydraulic tentacles around the mouth. They appear to be most closely related to mollusks.



GOLFINGIA VULGARIS
This species of peanut worm, found on northwest Atlantic coasts, lives in mud below the low-tide mark. It is 4 in (10 cm) long.

Phylum Bryozoa

Bryozoans

Occurrence c.6,000 spp. worldwide; attached to hard surfaces or aquatic environments

Also called sea mats, ectoprocts, or polyzoans, bryozoans are tiny aquatic animals, less than 1/32 in (1 mm) long, which live inside boxlike cases that are joined together to form colonies. Some bryozoans grow as encrustations on rocks and seaweeds, but others develop a branching, plantlike shape and wave backward and forward in the water. Once dead, these branching colonies are often washed up on the shore, where they may be mistaken for dried-out seaweed. This phylum is ecologically important as a source of food for some marine invertebrates.

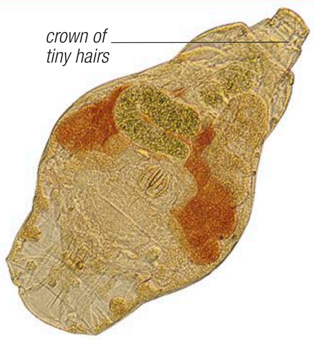
ROSS CORAL
Bryozoans bud continuously until thousands of interconnected "zooids" are produced. The North Atlantic Pentapora fascialis *may form colonies up to 6½ ft (2 m) in diameter.*

Phylum Rotifera

Rotifers

Occurrence c.2,000 spp. worldwide; in vegetation in lakes, rivers, and seas, microfreshwater habitats on land

Rotifers are among the smallest animals: the largest is just ⅛ in (3 mm) long. They live in water, and are either free-swimming, or attached to solid surfaces. The trumpet-shaped to spherical body usually has a "wheel organ" in front, formed by 2 whorls of hairlike structures (cilia), which is used for feeding and locomotion. Many are transparent, and several are parthenogenetic, males among these being unknown.



UNIDENTIFIED SP.
Although some rotifers are marine, most live in freshwater habitats, including temporary pools such as those in gutters and water films around leaves and soil particles. Like water bears (see opposite), they are specially adapted to survive drought. They can exist in a dried-out state for many years, coming back to life as soon as they are rehydrated.

Phylum Nemertea

Ribbon worms

Occurrence c.1,400 spp. worldwide; on bed and in surface or middle waters of seas, rivers, and lakes, in forests

A key feature of these mainly marine worms is their unique proboscis. This muscular tube, housed above the gut, is pushed out by hydraulic pressure in order to catch prey; in one large group of species, it bears piercing barbs that allow it to inject toxins. These worms range from less than 1/64 in (0.5 mm) in length to over 165 ft (50 m). All have a circulatory system and most have eyes—up to 250 in some species. Many are brightly colored. Two groups of these worms are commensal—one within the shells of bivalve mollusks, the other on and within the shell of a variety of crabs.

UNIDENTIFIED SP.
Ribbon worms are noted for their flat and sometimes extraordinarily long bodies. The worm shown here belongs to a group that includes the world's longest animal, which reaches a length of 180 ft (55 m). This is twice as long as the average adult blue whale.